MILITARY SPECIFICATION

RADIAC SET AN/PDR-45( )

1. SCOPE

1.1 This specification covers 1 type of portable, probe-type, battery-operated radiac detector computer-indicator which shall measure gamma radiation in the range up to 500 roentgens per hour in 3 decade scales and shall detect beta radiation. This equipment is for shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	- Box, Fiberboard
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing)

MILITARY

MIL-E-1/987	- Electron Tube, Gaseous Discharge, Type 7617
MIL-E-1/988	- Electron Tube, Regulator, Type 7615
MIL-B-18/9	- Battery, Dry, BA-30
MIL-P-116	- Preservation, Methods of
MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for (Navy)
MIL-D-3464	- Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-Q-9858	- Quality Program Requirements
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore: General Specification (Navy)
MIL-M-19590	- Marking of Commodities and Containers to Indicate Radioactive Material
MIL-B-55521	- Battery, Dry Packaging and Packing

STANDARDS

MILITARY

MIL-STD-108	- Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment
MIL-STD-129	- Marking for Shipment and Storage

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MIL-STD-147 - Palletized and Containerized Unit Loads, 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment
MIL-STD-415 - Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-781 - Reliability Tests: Exponential Distribution
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production
MIL-STD-810 - Environmental Test Methods

PUBLICATIONS

MILITARY
MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Ill. 60606).

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N. W., Washington, D. C. 20036).

DEPARTMENT OF TRANSPORTATION
Code of Federal Regulations - Specification 55, Title 49, Parts 170 to 190 and Title 14, Part 103

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. *REQUIREMENTS*

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

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3.2 General. The equipment shall be in accordance with all provisions of MIL-E-16400, except as otherwise specified herein.

3.2.1 Preliminary drawings. Preliminary drawings, for first article and quality conformance inspection, shall be prepared and submitted as specified (see 6.1.1).

3.3 Reliability.

3.3.1 Reliability program plan. The contractor shall prepare for approval a reliability program plan in accordance with this specification and MIL-STD-785. The reliability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the reliability program. The reliability program plan shall address the applicability or nonapplicability of each paragraph of MIL-STD-785. A preliminary reliability program plan shall be submitted as part of the technical proposal. A detailed reliability program plan shall be submitted to the command or agency concerned after contract award for Government approval (see 6.1.2).

3.3.2 Quantitative reliability requirement. The Radiac Set AN/PDR-45 ( ) shall have a 1000 hour specified (θ_o) mean-time-between-failure (MTBF).

3.3.3 Reliability prediction. A design reliability prediction shall be performed based on stress factors and part populations technique of MIL-HDBK-217. The prediction shall be based on the operational requirements of this specification. The reliability mathematical model and reliability block diagram shall be submitted with the reliability prediction report. Mechanical parts shall be included in the prediction. The source and applicability of all failure data used for the prediction shall be documented and justified prior to use. The results of the prediction shall be compared to the specified MTBF and where the predicted values do not achieve the requirements, the contractor shall accomplish such changes in design as are necessary to increase the predicted values sufficiently to achieve the requirements. A initial report shall be prepared for the command or agency concerned after contract award on the procedures, results, and conclusions of the reliability prediction (see 6.1.2).

3.3.4 Design review. The contractor shall conduct a design review to evaluate the equipment design and determine problem areas prior to the start of production. The command or agency concerned shall reserve the right to participate in the design review. Prior to the start of the design review, the command or agency concerned shall be notified for attendance of its representatives and provided with an advanced copy of the design review agenda. The results and recommendations of the design review shall be documented and submitted to the command or agency concerned for review and approval.

3.3.5 Reliability demonstration. The reliability of the Radiac Set AN/PDR-45 ( ) shall be demonstrated as specified in 4.2.1.

3.3.6 Reliability reports.

3.3.6.1 Letter reports. The contractor shall prepare a letter report describing the progress and results of all reliability effort performed during

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the reporting period and the reliability activities planned for the next reporting period. Reports shall be prepared after the end of the reporting period for the command or agency concerned. Reports shall be furnished for the duration of the contract.

3.3.6.2 Reports. The contractor shall prepare a report on the procedures, results, and conclusions for each of the tasks of 3.3.3, 3.3.4, and 3.3.5. These reports shall be submitted after completion to the command or agency concerned (see 6.1.2).

3.3.6.3 Failure reports. The contractor shall prepare a failure report whenever any equipment undergoing testing required by this specification fails. The failure reports shall contain all information necessary to determine the cause of failure and the corrective action required. The data to be reported shall be in accordance with the information requirements of Failure Identification Report of MIL-STD-781. Failure reports shall be submitted to the command or agency concerned (see 6.1.2).

3.4 Design requirements, general.

3.4.1 Environmental conditions. The equipment shall be capable of operation under the following conditions as specified in MIL-E-16400:

(a) Temperature - Class 3 (operating and nonoperating) (not including batteries)
(b) Humidity

3.4.1.1 In addition, the equipment shall operate after being subjected to pressures equivalent to any altitude up to 50,000 feet.

3.4.2 Electrical design. The equipment shall be battery powered.

3.4.3 Maintenance. Test points and test features are required in accordance with MIL-E-16400 and MIL-STD-415.

3.4.4 Mechanical.

3.4.4.1 Shock, vibration, inclination, and noise. Inclination and noise do not apply to this specification.

3.4.4.1.1 Shock. The shock requirements shall be grade A, class I, lightweight, type A, as specified in MIL-S-901. These requirements apply with the radiacmeter and probe assembled.

3.4.4.1.2 Vibration. The vibration requirement shall be type I of MIL-STD-167.

3.4.4.1.3 Calibration change. The calibration shall change no more than 20 percent as the result of shock and vibration tests.

3.4.5 Controls, indicators, and panel layout.

3.4.5.1 Tactile identification of controls is not required.

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3.4.5.2 All controls shall be classified as continually required, except calibration controls.

3.4.5.3 Elapsed time meters are not required.

3.4.5.4 Blown fuse indicators are not required.

3.4.6 Instruction plates. Instruction plates are not required, but required technical information shall be in the technical manual.

3.4.7 Mechanical security of external components. The probe, knobs, and other components on the exterior of the radiacmeter housing shall be secured to the housing by means which prohibit their working loose.

3.5 Performance and special requirements (principles of operation). The equipment shall detect and measure the rate of gamma radiation by means of a Geiger-Mueller Counter intermittently sensitized by the application of pulsed operating voltage. The sensitivity shall be adjustable to correspond to each scale range by appropriate modification of the pulsed operating parameters. The equipment shall compute and integrate the counts and present them in terms of roentgens per hour on an indicating microammeter whose reading shall be proportional to the applied radiation.

3.5.1 Detection and measurement.

3.5.1.1 Scale ranges. The equipment shall detect, measure, and indicate gamma radiation in the range 0 - 500 roentgens per hour, in 3 decade scales, as follows:

(a) 0 - 500 roentgens per hour
(b) 0 - 50 roentgens per hour
(c) 0 - 5 roentgens per hour

3.5.1.2 Accuracy. Subject to the conditions herein, the equipment shall measure and indicate gamma radiation with error not more than ± 20 percent of the true value of radiation intensity from any combination of errors introduced by the following:

(a) Battery voltage (see 3.6.6.1)
(b) Temperature (see 3.4.1)
(c) Humidity (see 3.4.1)
(d) Altitude (see 3.4.1.1)
(e) Calibration range (see 3.5.2.4)
(f) Shelf life (see 3.5.2.6)

3.5.1.2.1 Accuracy restriction. The stated accuracy shall be required only for the portions of the scales between 10 and 100 percent of full scale value.

3.5.2 Calibration.

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3.5.2.1 Calibration controls. The minimum number of controls shall be furnished to provide for complete adjustment of all 3 scales according to the requirements of 3.5.1.1. All calibration controls shall be accessible through a calibration control cover. The calibration controls shall be of such value that they are set at midrange when the instrument is calibrated and provide for at least a ± 20 percent variation on either side of the correct calibration value.

3.5.2.2 Calibration standards. Cobalt 60 or Cesium 137 shall be the standards of reference for gamma calibration. The calibration sources actually used shall be approved by the command or agency concerned.

3.5.2.3 Calibration method. For the purpose of calibration of the equipment, the probe shall be mounted so that its axis is perpendicular to the direction of radiation. Distances shall be measured to the "calibration center" of the probe as indicated by the circle on the probe.

3.5.2.4 Calibration settings. The range settings and radiation field intensities for calibration shall be as follows:

Selector switch	Desired meter reading	Co^{60} Field intensity	Cs^{137} Field intensity
500 50 5 r/hr	400 40 4 r/hr ± 2 percent	400 40 4 r/hr ± 2 percent	472 47.2 4.72 r/hr ± 2 percent

Two additional readings shall be made on each scale, at approximately 40 percent and 15 percent of full scale value.

3.5.2.5 Directionality. The readings shall be within the allowed tolerance (see 3.5.1.2) for all orientations of the probe up to 45 degrees from the calibrating orientation (see 3.5.2).

3.5.2.6 Stability (shelf life). Within a 30-day period following calibration, no reading shall change more than 10 percent from its reading at calibration, nor progress beyond the allowable calibration accuracy limits (see 3.5.1.2).

3.5.2.7 Energy dependence. Within the intensities specified in 3.5.1.2.1, the response shall be within ± 20 percent of the true value over the gamma energy range between 0.08 and 2 million electron volts (MeV). Directional orientation shall be as specified in 3.5.2.5.

3.5.3 Circuit response time. The circuit time constants shall be such that for a deflection of 20 percent or more of full scale, the time for the meter to indicate 90 percent of true value shall be between 5 and 10 seconds.

3.5.4 Saturation effects. When the radiacmeter is subjected to gamma radiation of intensity higher than the maximum value of any scale in use, up to 10,000 roentgens per hour, it shall indicate off-scale readings at the high end of the scale.

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3.5.5 Noise and spurious indications. No residual or spurious counts shall be introduced by circuitry except from switching.

3.5.6 Beta response. The equipment shall detect and indicate beta radiation for energies above 0.1 MeV, with the probe window open, but will not necessarily be calibrated for beta radiation.

3.6 Equipment composition. The equipment shall include the following component parts:

(a) Radiacmeter
(b) Probe
(c) Radioactive test sample
(d) Carrying case

3.6.1 Radiacmeter and probe assembly. The radiacmeter and probe as an assembly shall be designed for convenient portability and balance. The meter and controls shall be located on the top of the radiacmeter housing. The probe and radiacmeter shall be joined by a plug-in connector.

3.6.1.1 Degree of enclosure. The degree of enclosure shall be watertight in accordance with MIL-STD-108 with the probe attached to the radiacmeter housing and the beta shield closed.

3.6.2 Dimensions and weights. The maximum dimensions and weights shall be:

Radiacmeter:	9-1/2 inches height
	5-1/4 inches width
	8-1/4 inches depth
	4 pounds - weight
Probe:	1-3/8 inches width
	1-1/4 pounds - weight

The length of the probe shall be such that when attached to the radiacmeter housing, the distance from the detector to the indicating meter shall be at least 41 inches.

3.6.3 Radiacmeter. The radiacmeter shall be adapted to receive the plug-in type probe. The radiacmeter shall include the following circuitry:

(a) Stable pulse rate computer
(b) Indicating meter
(c) High voltage power supply
(d) Pulse generator

3.6.3.1 Assembly. The several subassemblies and components of the radiacmeter circuit shall be mounted on a circuit board which shall be supported by the top of the radiacmeter housing. The requirement for accessibility of components shall not be compromised.

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3.6.3.2 Range switch. There shall be a 5-position selector switch mounted on the top of the radiacmeter housing, which shall select the positions and ranges as follows:

(a) OFF
(b) BATTERY CHECK
(c) 500 roentgens per hour
(d) 50 roentgens per hour
(e) 5 roentgens per hour

The housing shall be marked to indicate each selector switch position.

3.6.3.3 Illumination switch. There shall be a momentary-contact switch to control the illumination lamp.

3.6.4 Pulse rate computer. The pulse rate computer shall be essentially a bi-stable multivibrator circuit feeding the indicating meter.

3.6.5 Indicating meter. The indicating meter shall be a 500-microampere type, with face marked in 1 scale. The meter pointer shall not be subject to deviation by reason of electrostatic charges on the meter window. The meter shall be mounted on the top panel.

3.6.6 Battery power requirements.

3.6.6.1 Battery. The power source shall be a battery consisting of two type BA-30 cells in accordance with MIL-B-18/9 (commercial size D). The battery is not required to be furnished with the equipment.

3.6.6.2 Battery drain. At room temperature the radiacmeter, with the illuminating lamp OFF, shall draw no more than 120 milliwatts from the battery. With the illuminating lamp ON, the total power drain shall be no more than 315 milliwatts. Battery end point shall be 1.8 volts.

3.6.6.3 Power supply. The power supply shall furnish the necessary operating voltages from the battery input. The high voltage regulation circuit shall include the corona regulator tube type 7615 in accordance with MIL-E-1/988.

3.6.6.4 Pulse generator. The pulse generator shall furnish the necessary pulses to the probe to enable the Geiger-Mueller tube to detect radiation over a particular duty cycle. The pulse generator circuit shall include the relaxation oscillator tube type 7617 in accordance with MIL-E-1/987.

3.6.7 Meter illumination. The meter shall be illuminated by means of an incandescent lamp, with optical accessories as required, to apply uniform white illumination over the entire scale. The lamp shall be controlled by a push-button type switch located on the top panel. The lamp shall draw not more than 195 milliwatts from the battery. The illumination shall be sufficient for reading the meter under dim, but not dark adapted, ambient conditions.

3.6.8 Radiacmeter housing. The housing shall be designed for easy and convenient use during survey operations, with the smallest dimensions practicable.

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Form factor shall be subject to approval of the command or agency concerned. The housing shall be protected against leaks through the openings for external operating controls, hold-down screws, and other openings, by means of O-ring seals or the equivalent. No drainage plug shall be provided. The housing shall be of split type construction, using a gasketed seal, and held together by recessed captive screws from the top surface. There shall be a receptacle on the front end for attaching the probe.

3.6.8.1 Radiacmeter housing compartmentation. The housing shall be divided into separate watertight compartments for the chassis and for the batteries, respectively. There shall be separate access to the battery compartment to permit changing batteries without opening the chassis compartment. Electrical connections between the compartments shall be by means of insulated feed-through connectors through the battery compartment wall, permanently attached to the wiring. The battery compartment and connectors shall not permit leakage of battery fluids into the chassis compartment.

3.6.8.2 Radiacmeter housing materials.

3.6.8.2.1 Metal. The housing shall be constructed of aluminum as light as possible consistent with the strength required. The protection of aluminum (anodizing) shall be in accordance with MIL-E-16400.

3.6.8.2.2 Paint. Paint shall be in accordance with MIL-E-16400 for portable shipboard equipment.

3.6.8.2.3 Housing surfaces. All surfaces shall be as smooth as practicable, with a minimum number of sharp edges, crevices, or other surface irregularities to collect contaminated material. Indication markings engraved on the housing shall be filled with nonporous paint of contrasting color so as to produce a smooth surface. Rubber or felt feet shall not be used. Parts subject to contamination shall be readily capable of decontamination without recourse to laboratory equipment or techniques.

3.6.8.2.4 Handcarrying handle. The radiacmeter, subject to the approval of the command or agency concerned, shall be equipped with a fixed handle for carrying the instrument in 1 hand. The balance point of the radiacmeter with probe shall be at the center of the handle. The requirements as to smooth surfaces apply also to the handle.

3.6.9 Probe. The probe shall consist of a housing for the Geiger-Mueller detector tube, mounted on a semirigid support. The detector housing and support shall be 1 continuous unit. All surfaces of the probe shall conform to the requirements of 3.6.8.2.3.

3.6.9.1 Detector housing. The detector housing shall consist of a cylindrical metal enclosure for the Geiger-Mueller detector tube, with a beta window and movable beta shield located axially at the outer end. The detector housing shall also enclose the energy-correcting shield. The housing shall be compatible with the radioactive test sample (see 3.6.10).

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3.6.9.2 Probe support. The probe support shall be a semirigid tubular structure to provide mechanical support and electrical connections between the detector housing and the radiacmeter housing. The probe support, when bent, shall retain its configuration until bent into another configuration. The probe support shall not be damaged by repeated bending into circles or arcs of 7-inch inside diameter.

3.6.9.3 Calibration marking. A circle shall be engraved around the circumference of the probe housing at a position corresponding to the most sensitive part of the Geiger-Mueller tube. The center of the circle shall be designated the calibration center.

3.6.10 Radioactive test sample. A test sample containing a radioactive source shall be provided to check the overall functioning of the equipment. The sample shall consist of a mechanical assembly designed to fit over the end of the probe and apply beta radiation from the source to the Geiger-Mueller tube.

3.6.10.1 Radioactive source. The radioactive source in the test sample shall consist of a capsule of Krypton 85 gas. The source shall be constructed, sealed, and mounted in accordance with Specification 55, Title 49, Parts 170 to 190 and Title 14, Part 103 of the Code of Federal Regulations.

3.6.10.2 Mounting. The source shall be mounted in a sliding holder which shall be automatically retractable into a shielded housing. When the sample is in place on the probe end, the source and openings shall be aligned to apply beta radiation to the beta window of the Geiger-Mueller tube. When the source is retracted, there shall be no beta emission from the sample.

3.6.10.3 Intensity. The intensity of the source shall be adjusted to cause a deflection of 4.0 roentgens per hour, ± 10 percent, in the equipment meter.

3.6.10.4 Loss prevention and storage. The test sample shall be attached to the carrying case by a chain which may be extended 24 inches from the case. The strength of attachment shall be sufficient to prevent unintentional detachment of the sample from the case by a force of less than 60 pounds. When not in use, the sample shall be stored in the sample storage compartment of the carrying case. The method of attachment and the storage compartment shall be subject to approval by the command or agency concerned.

3.6.10.5 Markings. In addition to the required markings for radioactive material (see 3.8), the test sample shall include a serial number which will be assigned by the command or agency concerned.

3.7 Carrying case. The equipment shall include a carrying case designed to hold the following:

(a) Radiacmeter
(b) Probe
(c) Test sample
(d) 2 technical manuals
(e) 2 spare battery cells

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Means shall be provided to secure all items in the carrying case to prevent shifting and damage during shipment and handling. The design of the carrying case shall be subject to approval of the command or agency concerned.

3.7.1 Construction.

3.7.1.1 Material. The material shall be drawn aluminum alloy. Rubber feet shall not be used. The surface shall be as smooth and free of crevices as practicable to permit removal of contaminants.

3.7.1.2 Form. The carrying case shall provide storage for the probe when coiled into a circular configuration not larger than 10-inch outside diameter. It shall not be necessary to reduce the diameter of the coil of the probe below 9 inches for insertion or removal from the case. The cover shall be the outside-fitting type. Overall shape shall permit convenient vertical stacking. There shall be 1 handle located on the right side.

3.7.1.3 Compartment for test sample. A compartment shall be provided in the carrying case to store the test sample and chain. The compartment shall open at the top with the cover designed to be held in closed and open positions by means of a detent service. The compartment shall be conveniently accessible from the top of the carrying case for 2-hand manipulation of the contents (see 3.6.10.4).

3.7.1.4 Degree of enclosure. The degree of enclosure shall be splash-proof in accordance with MIL-STD-108.

3.7.1.5 Maintainability. The equipment shall be constructed to provide ease of maintenance, accessibility, and replacement of all parts and components.

3.8 Marking for radioactive material. The marking for radioactive material on the test sample compartment and carrying case shall be in accordance with MIL-M-19590. On the carrying case, the radioactivity marking shall be located on the same side as the identification plates.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement. The contractor shall provide and maintain a written quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858 (see 6.1.2).

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4.1.1.1 Quality assurance plan. A quality assurance plan applicable to the equipments covered by this specification shall be prepared for Government approval (see 6.1.2).

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause of suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Tests, test equipment, and test procedures. The contractor shall submit a list of proposed tests and written detailed test procedures, including test equipment and tolerance limits to be used for approval (see 6.1.2). Test procedures shall be identified in accordance with the classifications in 4.2.

4.2 Classification of inspection. The method of examination and testing of the equipment shall be classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)

(c) Reliability testing (see 4.5)
(d) Inspection of preparation for delivery (see 4.6)

4.2.1 Reliability demonstration. The reliability requirements of 3.3.1 shall be demonstrated in accordance with the requirements of MIL-STD-781. Reliability testing will be performed at test level C with test plan III for the initial production units. Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the specification. Testing shall be performed on 6 or more production units until sufficient test time (in accordance with the paragraph on test length in MIL-STD-781) permits either an accept or reject decision in accordance with the accept-reject criteria for test plan III of MIL-STD-781. Preventative maintenance shall not be allowed during reliability demonstration testing. The tested equipment shall be returned to production for reinspection and reconditioning, as required, and may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.3 First article inspection. Unless otherwise specified (see 6.1.1), 3 units shall be required for first article inspection. First article inspection shall consist of the examinations and tests specified in 4.3.1.

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4.3.1 First article inspection shall be as specified in MIL-E-16400, except that the following tests shall not apply:

(a) Supply line voltage and frequency
(b) Power
(c) Water cooling
(d) Accelerated life
(e) Inclination and noise
(f) Reliability

4.3.1.1 The following tests shall be added:

(a) Battery power (see 3.6.6.1)
(b) Accuracy (see 3.5.1.2)
(c) Energy dependence (see 3.5.2.7)
(d) Saturation (see 3.5.4)
(e) Circuit response time (see 3.5.3)
(f) Altitude (Method 500, Procedure I of MIL-STD-810)

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be in accordance with MIL-E-16400 and, in addition, shall include the following: calibration for gamma at 3 points on each scale in accordance with 3.5.2.4; check for beta on each scale (see 3.5.6).

4.4.1.1 Prior to production inspection, each equipment shall be operated for 72 hours in the counting mode, 24 hours on each scale range. For this purpose, the test sample may serve as the source, and external power applied.

4.4.2 Production control inspection. Production control inspection shall be in accordance with MIL-E-16400, except that the following tests shall not be performed:

(a) Supply line voltage and frequency
(b) Power
(c) Water cooling
(d) Weld test
(e) Radio interference and undesired radiation

4.4.2.1 Battery power (see 3.6.6.2) and the accuracy test to determine conformance with 3.5.1.2 shall be included in the production control inspection.

4.4.2.2 Sampling for production control inspection. Samples for production control inspection shall be selected at random from those equipments which have passed the production inspection (see 4.4.1.1). The sample size shall be determined in accordance with Table I.

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Table I. Sampling for production control inspection.

Number of radiac sets in inspection lot	Number of radiac sets in sample	Number of equipments nonconforming on any tests	
		Acceptance number	Rejection number
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 and over	15	1	2

4.4.2.2.1 Each of the sample radiac sets selected in accordance with 4.4.2.2 shall be subjected to each of the tests specified in 4.4.2 to determine compliance with this specification. Failure to conform shall be counted as a defect and the radiac set shall be rejected. If the number of such nonconforming radiac sets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Reliability testing. The reliability of the equipment shall be demonstrated and reported in accordance with MIL-STD-781. The specified MTBF for the reliability tests shall be 1000 hours.

4.5.1 Reliability (demonstration) phase of production reliability tests. The demonstration phase of reliability production testing shall be conducted on the initial production units; testing shall be performed at test level E with test plan V. The sample size shall be in accordance with MIL-STD-781 for the initial production month. The reliability requirements must be successfully demonstrated prior to the shipment of any production equipments.

4.5.2 Production acceptance (sampling) phase of production reliability tests. The production acceptance (sampling) phase of reliability tests shall be conducted at test level C with test plan VI. The sample size shall be in accordance with MIL-STD-781 for each month's production. The reliability test must be successfully demonstrated each month prior to shipment of production equipment for that month.

4.5.3 Reliability test plan. The contractor shall prepare a detailed reliability test plan in accordance with MIL-STD-781. Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the equipment specification. Performance parameters to be measured during the reliability test shall be in accordance with the equipment specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the Government. No preventative maintenance shall be performed during the reliability test. Prior to the start of reliability testing, the contractor shall submit a reliability test plan to the command or agency concerned for Government approval (see 6.1.2).

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4.5.3.1 Performance parameters. The performance parameters to be monitored shall include, but not be limited to, the following:

(a) Ranges (see 3.5.1.1)
(b) Accuracy (see 3.5.1.2)
(c) Circuit time constant (see 3.5.3)
(d) Power supply voltage (see 3.6.6.3)

4.5.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. All failures shall be considered relevant unless determined by the command or agency concerned to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service.

4.5.5 Final restoration and disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning as required, and may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.5.6 Reliability test reports. The contractor shall submit test reports for reliability testing in accordance with MIL-STD-781. Reliability test reports shall be prepared for Government approval (see 6.1.2).

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with section 5. First article packs which contain Level A packages shall be subjected to rough handling tests (see 5.1.1).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5.1.1.3 Preservative application. None.

5.1.1.4 Unit packaging. Each radiac set shall be individually packaged in accordance with Method II b of MIL-P-116, ensuring compliance with the General paragraph under Methods of Preservation (Unit Protection) and the Physical

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Protection paragraph therein and shall meet the rough handling tests of MIL-P-116. Desiccant shall conform to MIL-D-3464, Type II or III. Technical manuals shall be packaged in a waterproof, transparent, heat sealed bag, minimum 4-mil thick, and shall not be placed within the barrier used to package any other items. Cushioning and dunnage shall be used to prevent physical or mechanical damage. When applicable, batteries shall be packaged separately in accordance with MIL-B-55521.

5.1.2 Level C. Each complete radiac set shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1.1).

5.2.1 Level A. Radiac sets, packaged as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, V Grade, Special Use Requirement, for Type 2 Load. In lieu of the closure and waterproofing requirements in the appendix to PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, corners, and manufacturer's joint with waterproof tape, 2-inches minimum width, conforming to PPP-T-60, Class 1, or PPP-T-76. Reinforcing shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. Radiac sets, packaged as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, Class weather-resistant. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. Items, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier and which will ensure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Unitized loads. When applicable, items packed as specified in 5.2, shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to 1 destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping containers specified in 5.2. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.4 Marking. In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129 (see 6.1.1). Marking shall include special marking for Method II packs.

5.5 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Requirements for preliminary drawings (see 3.2.1).
(c) Number of first article samples if other than that specified in 4.3.
(d) Levels of preservation, packaging, and packing, and applicable marking (see section 5).

6.1.2 Contract data requirements. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include but are not restricted to the following:

(a) Preliminary drawings (see 3.2.1)
(b) Reliability program plan (see 3.3.1)
(c) Reliability prediction (see 3.3.3)
(d) Reports, as specified (see 3.3.6.2 and 3.3.6.3)
(e) Technical manuals are to be provided with each equipment if required in the contract or order (see 3.7)
(f) Quality program (see 4.1.1)
(g) Quality assurance plan (see 4.1.1.1)
(h) Test procedures (see 4.1.3)
(i) Reliability test plan (see 4.5.3)
(j) Reliability test reports (see 4.5.6)

6.2 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Changes from previous issue. The outside margins of this document have been marked "#" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - EC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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